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9. What is really the moral of Lewis Carroll's bit about Achilles and the Tortoise?

In his short piece *What the Tortoise Said to Achilles*, Lewis Carroll uses a fictional dialogue between Achilles and a tortoise to illustrate a problem that lies at the foundations of logic.[1] The story begins with Achilles presenting two premises from Euclidean geometry: (A) things that are equal to the same are equal to each other, and (B) the two sides of this triangle are equal to the same. He attempts to conclude (Z) that the two sides of the triangle are equal to each other. The Tortoise agrees with A and B but refuses to accept Z until Achilles includes an additional rule connecting them: (C) if A and B are true, then Z must be true. When Achilles adds C, the Tortoise insists that he includes a further statement to connect A, B, and C to Z, and so on. The Tortoise guides Achilles through a seemingly never-ending series of steps to demonstrate to him an interesting regress occurring when closely examining the structure of logical arguments.

The regress explained in this story reflects the structure of Zeno's paradox of Achilles and the Tortoise, which Carroll is explicitly drawing from. In Zeno's version, Achilles can never overtake the Tortoise because he must first reach the point where the Tortoise began, then the point where the Tortoise moved next, and so on. Each finite step is clear, but when put together, it appears that motion as a whole becomes impossible given the endless number of subdivisions that can be made. Carroll applies this same structure to reasoning in this story to the motion of logical arguments; if every inference requires an additional premise to justify it, and each justification itself requires another, then logical motion going from premises to conclusion can never actually begin. The Tortoise accepts the premises and even the general principle, but he refuses to take the step of making an inference to the conclusion based on the premises. What Carroll shows is that inference does not follow automatically from accepting statements as true. It requires a different kind of commitment, and that is a step that is not captured by simply adding another proposition but by accepting and using a rule.

Carroll's point in this story is that simply accepting a list of true statements, even logical or mathematical ones, is not enough to guarantee that the reasoning between them will actually occur. The Tortoise accepts each premise and even the conditional that should link them to the conclusion, yet he withholds the conclusion because he treats each step in the inference as something that must be independently accepted. The Tortoise's refusal shows that logical inference cannot be reduced to a set of conventions or principles unless those conventions also act as rules. That is, unless the convention provides us with a way to apply it in specific cases, it can't justify inference. In other words, we can't explain logical reasoning just by listing additional accepted truths; we need something that does the work of taking us from one truth to the next. Since each new step requires its own convention applied uniquely to its

[1]Lewis Carroll (April 1895). "What the Tortoise Said to Achilles". Mind. IV (14): 278–280.

particular premises, and we are limited to finitely many conventions, then we are also limited to finitely many logical truths.

We imagine that logic is supposed to support infinitely many inferences. In the dialogue, "it follows" is treated like a missing convention that must be authorized repeatedly. What is missing are the rules of inference that permit you to go from one statement to another. These rules are not just further propositions; they're operations that act on the propositions. For example, universal instantiation is an operation that lets you apply a generalization to a particular case: from "All A's are B's" and "this is an A," you can infer "this is a B." Similarly, modus ponens allows you to go from a conditional, "If A then B", and the antecedent "A", to the conclusion "B." These are not extra facts that Achilles is writing into his notebook; they are what let logic move. A principle like "things equal to the same thing are equal to each other" can be accepted as true, but as long as we are lacking a rule for how to apply it, it remains inert and stands solitary next to the premises. What Carroll's dialogue points out is that inference itself is its own kind of step, and without rules that govern how those steps work, logic would never be able to make any progress.

This message touches on broader themes in 20th-century philosophy of mathematics. A. J. Ayer, a logical positivist, argued that it is not necessary to reduce everything to pure logic. He believed that some things can be accepted as true by convention. If this is in fact the case, then logical truths hold because of how we define terms and set up our systems, not because they reflect any deeper necessity. However, Carroll's story complicates this: if every step in a logical argument needs to be accepted as a separate convention, then we'd need an infinite number of conventions to justify even simple conclusions. A finite set of conventions could not be enough to carry us all the way from premises to conclusions in every case since there are infinite cases. Thus, holding certain statements true by convention isn't sufficient. To supplement this, we need rules that actually do the work of moving us from one step to the next.

Similarly, Saul Kripke's interpretation of Wittgenstein presents the way by which this becomes a problem about rule-following: how do we know we're applying a rule correctly, and what justifies that feeling of correctness? The Tortoise's refusal to make the leap from principle to conclusion illustrates that question. In logic, it seems that it is not enough to say "this is like that." You need a rule that lets you actually treat it as such.

Carroll's dialogue leaves us with a lasting moral: logical reasoning depends not just on what we accept as true, but on our capacity to move between truths. Without rules of inference, even the most basic steps cannot be made. The Tortoise's refusal to infer exposes the limits of treating logic as a collection of conventions or statements. What matters is not only agreement, but also the ability to do the logic and actually make the inference. The moral of the story, then, is that the ability to reason requires not just what we believe, but a way of following through on what those beliefs imply.